Formula auditing in Excel is a set of tools used to display, trace, and troubleshoot the relationships between cells and formulas. These tools are essential for verifying the accuracy of complex engineering models or financial reports like a **Salary statement**.

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1. Visualizing Relationships

Excel provides visual cues to help you see how data flows through your worksheet:

- **Trace Precedents**: Draws arrows showing which cells provide data to the currently selected formula.
- **Trace Dependents**: Draws arrows showing which cells rely on the value of the currently selected cell.
- **Remove Arrows**: Clears all tracer arrows from the worksheet once you have finished your audit.

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2. Troubleshooting Tools

When a formula returns an unexpected result or an error like `#VALUE!`, these tools help pinpoint the cause:

- **Error Checking**: Scans the worksheet for common formula errors and suggests potential fixes.
- **Evaluate Formula**: Allows you to step through a complex formula, such as a **Nested IF** or **INDEX and MATCH**, one part at a time to see how Excel calculates the final result.
- **Trace Error**: If a cell displays an error, this tool draws arrows to the source of the problem, which is particularly useful for finding the origin of a `#N/A` error in a **Lookup function**.

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3. Transparency and Documentation

For large-scale data management involving up to **1,048,576 rows**, maintaining transparency is key:

- **Show Formulas**: Toggles the display so you see the actual formulas in every cell instead of the calculated results. This is helpful for printing a "logic map" of your sheet.
- **Watch Window**: A floating window that lets you monitor the values of specific "target" cells (like a Grand Total) even while you are working on a different sheet or a distant row.
- **Comments and Notes**: Use these to document the logic behind complex calculations for other users.

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4. Auditing Workflow

1. **Select the Cell**: Highlight the formula you want to investigate.
2. **Trace Precedents**: Use the **Formulas** tab to see where the data is coming from.
3. **Evaluate**: If the result is wrong, use **Evaluate Formula** to watch the calculation step-by-step.
4. **Check Dependents**: Ensure that changing this cell won't unexpectedly break other parts of your **Pivot Table** or **Charts**.